

Micro Set 5. Public Goods & Information Failure — Model Answer Scheme

Question 1

[2 marks]

A **pure public good** is a good that is non-rivalrous (one person's consumption does not reduce the quantity available to others) and non-excludable (no one can be prevented from consuming it once provided). *[1 mark.]*

Evidence from Extract 1.1: (i) non-excludable — “no citizen or resident within Singapore can be excluded from the protection”; (ii) non-rivalrous — “one resident's enjoyment of protection does not diminish another's.” *[1 mark for both features identified.]*

Examiner Note. Both features must be named with examples. Mentioning only one (e.g. “free for all”) earns 1 mark.

Question 2

[2 marks]

Used-car lemon-law complaints rose persistently between 2019 and 2023, from 1840 to a peak of 2680 (+46%), before easing slightly to 2520 in 2024. The complaint rate per transaction rose more sharply, because the volume of transactions fell from 79100 to 54400 over the same period. *[1 mark for direction + magnitude, 1 mark for transaction-rate insight.]*

Question 3

[4 marks]

Why free market fails to provide national defence.

National defence is non-excludable (Extract 1.1). This means that once provided, every resident benefits regardless of whether they paid. *[1 mark.]*

This generates the *free-rider problem*: rational individuals have an incentive to under-state their true willingness to pay, anticipating that others will fund the good while they consume it freely. The aggregate revealed demand in a hypothetical market would be far below true aggregate willingness-to-pay (true MSB). *[2 marks for free-rider mechanism.]*

Even if every individual privately valued defence at, say, S\$100 per year, no one has the incentive to pay the first dollar unless others have committed first. The market therefore fails to coordinate the contributions necessary for adequate provision. In the limit, free-market provision approaches zero, far below the socially optimal Q^* . “It cannot work otherwise,” as the Ministry of Defence states (Extract 1.1). *[1 mark for under-provision conclusion.]*

Examiner Note. The free-rider problem must be explicitly named and explained. “Nobody wants to pay” alone is insufficient — the analytical content is the rational incentive structure given non-excludability.

Question 4

[4 marks]

Asymmetric information and the lemon problem.

In the used-car market, sellers possess private information about the vehicle's true condition, accident history, and maintenance state, while buyers cannot easily verify these characteristics at the point of sale. *[1 mark for asymmetric information definition.]*

Because buyers cannot distinguish high-quality from low-quality cars before purchase, they are willing to pay only an *average* price reflecting the average expected quality across the market. Sellers of genuinely high-quality cars find this average price below their reservation price (the true value of their car), and withdraw from the market. *[2 marks for adverse-selection mechanism.]*

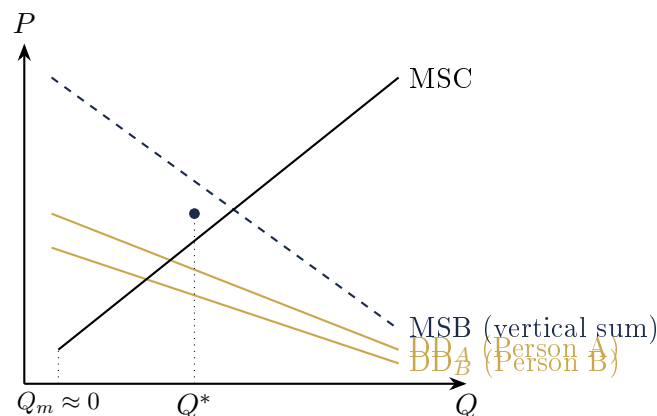
As high-quality sellers withdraw, the remaining market shifts toward lower-quality cars on average. Buyers' average expected quality falls, and so does the price they are willing to pay — triggering further withdrawal of medium-quality sellers. The market for high-quality used cars unravels. The empirical signal of this dynamic is rising lemon-law complaints (1840 in 2019 to 2680 in 2023, Extract 1.1) despite falling transaction volumes (79100 to 60200). *[1 mark for evidence and unravelling.]*

Examiner Note. The key analytical sequence: asymmetric info → average pricing → adverse selection → market unravels. All four steps are necessary for full marks.

Question 5

[6 marks]

Why the free market under-allocates resources to pure public goods.



Unlike private goods, the market demand curve for a public good is the *vertical sum* of individual demand curves (DD_A , DD_B , etc.) at each quantity, because all consumers benefit from the same unit simultaneously (non-rivalry). The vertically summed curve is the social marginal benefit (MSB). The socially optimal output occurs where $MSB = MSC$ at Q^* . *[2 marks for diagram + vertical-sum logic.]*

However, the free market cannot reach Q^* . In a private market, each consumer compares their own MPB (DD_A or DD_B individually, not the sum) with the price, and chooses individually. With non-excludability, no consumer would voluntarily reveal their true willingness-to-pay; rational free-riding leads to revealed demand approaching zero. The market-determined output $Q_m \approx 0$, far below Q^* . *[2 marks for free-rider → $Q_m \approx 0$.]*

The under-allocation $|Q^* - Q_m|$ represents the welfare loss from market failure. The government must therefore step in to provide the public good directly, funded by taxation, which mandates the contribution and eliminates the free-rider incentive. This is precisely the framework Singapore uses for defence (S\$19.8 billion in 2024, Table 1.1), funded universally through taxation. *[2 marks for government-provision conclusion + data anchor.]*

Examiner Note. The vertical-sum demand curve is the crucial diagrammatic feature for public-good analysis. Drawing horizontal-sum (the private-good convention) signals fundamental misunderstanding of non-rivalry.

Question 6

[6 marks]

Why information disclosure may be more efficient than regulation.

Reason 1: Targets the root cause. The lemon problem arises from *information asymmetry*, not from sellers' market power. Direct regulation (e.g. mandatory price controls, mandatory dealer licensing) addresses symptoms but not the cause. The One Motoring portal directly remedies the asymmetry: it provides buyers with the inspection records, accident history, and registration details that previously only the seller possessed. With symmetric information, the market can self-correct. [2 marks.]

Reason 2: Preserves market efficiency and choice. A regulatory approach (e.g. a Lemon Law) requires costly enforcement, generates compliance costs for dealers, and can lead to distortions where dealers withdraw from the market entirely if regulatory burden is too high. The disclosure approach preserves all transactions while enabling buyers to make informed decisions, allowing the market to clear at quality-differentiated prices. "Disclosure is more efficient than regulation when the underlying problem is information" (Extract 1.1). [2 marks.]

Reason 3: Complements rather than replaces existing mechanisms. The Lemon Law remains in place as a backstop for cases where information disclosure fails (e.g. deliberate misrepresentation by sellers). The combination of ex-ante information disclosure (One Motoring) and ex-post legal remedy (Lemon Law) is more efficient than either alone, addressing different stages of the transaction. The CASE complaint data (Extract 1.1) shows the Lemon Law has "reduced but not eliminated" the problem, reinforcing the need for information-based complements. [2 marks.]

Examiner Note. The strongest answers frame disclosure and regulation as *complements* attacking different points in the information-transaction chain — not as substitutes.

End of Set 5 — Model Answers